project proposal

1. Set up

Packages

```r
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# read .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)

library(dplyr)     # data manipulation
library(broom)     # tidy statistical output
library(stringr)   # distribution plot
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")


# read in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# read in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

2. Cleaning

`sites` object

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

Cleaning `taxon_list`

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

Cleaning `water_quality`

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(dissolved_oxygen_mg_l, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

Cleaning `aq_ins`

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
```

```r
        diptera_chironomid,
        annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_do, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

Focus Taxa

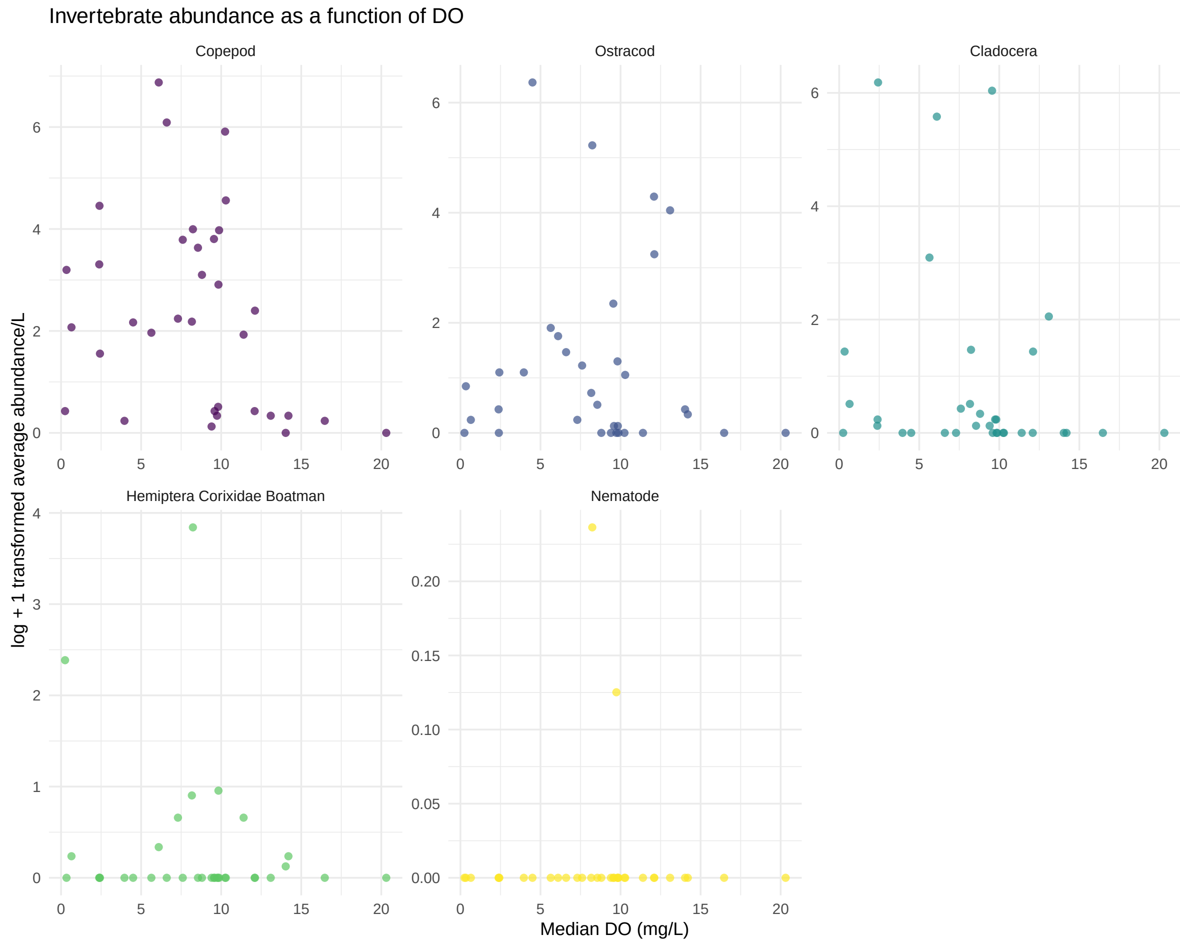
```r
# taxa to include in all figures
focus_taxa <- c("Ostracod", "Copepod", "Cladocera",
                "Hemiptera Corixidae Boatman", "Nematode")
```

Figure 1: Invertebrate abundance as a function of DO

```
# create object DO_taxa from aq_ins_clean
DO_taxa <- aq_ins_clean |>
  # remove rows that are missing median salinity values
  filter(!is.na(med_do)) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # mutates to title-case taxon names
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  # only include focus taxa
  filter(taxon %in% focus_taxa) |>
  # order taxon levels by maximum average abundance per liter
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
  # keep only the columns needed for the figure
  select(date, site, site_full, taxon, ave_lit, log_ave_lit, med_do)
```

```
# base layer: ggplot
DO_taxa_plot <- ggplot(data = DO_taxa, # assigns plot to object do_taxa_plot
                        # uses sal_taxa data frame
                        mapping = aes(x = med_do, # x-axis: med_do
                                      y = log_ave_lit, # y-axis: log_ave_lit
                                      color = taxon)) + # color by taxon
  # first layer: points show individual abundance measurements
  geom_point(shape = 16, # shape is circle
             size = 2, # size is 2
             alpha = 0.7) + # opacity is 0.7
  # separate panels for each taxon with free x- and y-axis scales
  facet_wrap(~ taxon, scales = "free") +
  # use non-default colors for points
  scale_color_viridis_d() +
  labs(x = "Median DO (mg/L)", # relabel x-axis
       y = "log + 1 transformed average abundance/L", # relabel y-axis
       # creates descriptive title
       title = "Invertebrate abundance as a function of DO") +
  # uses non-default theme
  theme_minimal() +
  # remove redundant legend
  theme(legend.position = "none")

# shows plot
```

```
DO_taxa_plot
```



shows that individual taxa have noisy, weak, or inconsistent relationships with DO.

Figure 2: Aquatic Invert. Abundance Across Sites

```
# creating new object from clean aquatic inverts
abundance_by_site <- aq_ins_clean |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # convert taxon to title case
  mutate(taxon = str_replace_all(taxon, "_", " "),
```

```r
         taxon = str_to_title(taxon)) |>
  # filter to include taxa of interest
  filter(taxon %in% focus_taxa)
```

```r
# base layer

abundance <- ggplot(data = abundance_by_site,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   # color by taxon
                   mapping = aes(x = site_full,
                                 y = log_ave_lit,
                                 color = taxon)) +
  # first layer: jitter points to represent observations
  geom_jitter(width = 0.15,
              height = 0,
              alpha = 0.7) +
  # facet by taxon
  facet_wrap(~ taxon) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # non-default colors
  scale_color_viridis_d() +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Aquatic Invert. Abundance Across Sites") +
  theme_bw() +
  # remove legend
  theme(legend.position = 'none',
        # fix x-axis for readability
        axis.text.x = element_text(angle = 45, hjust = 1))

abundance
```

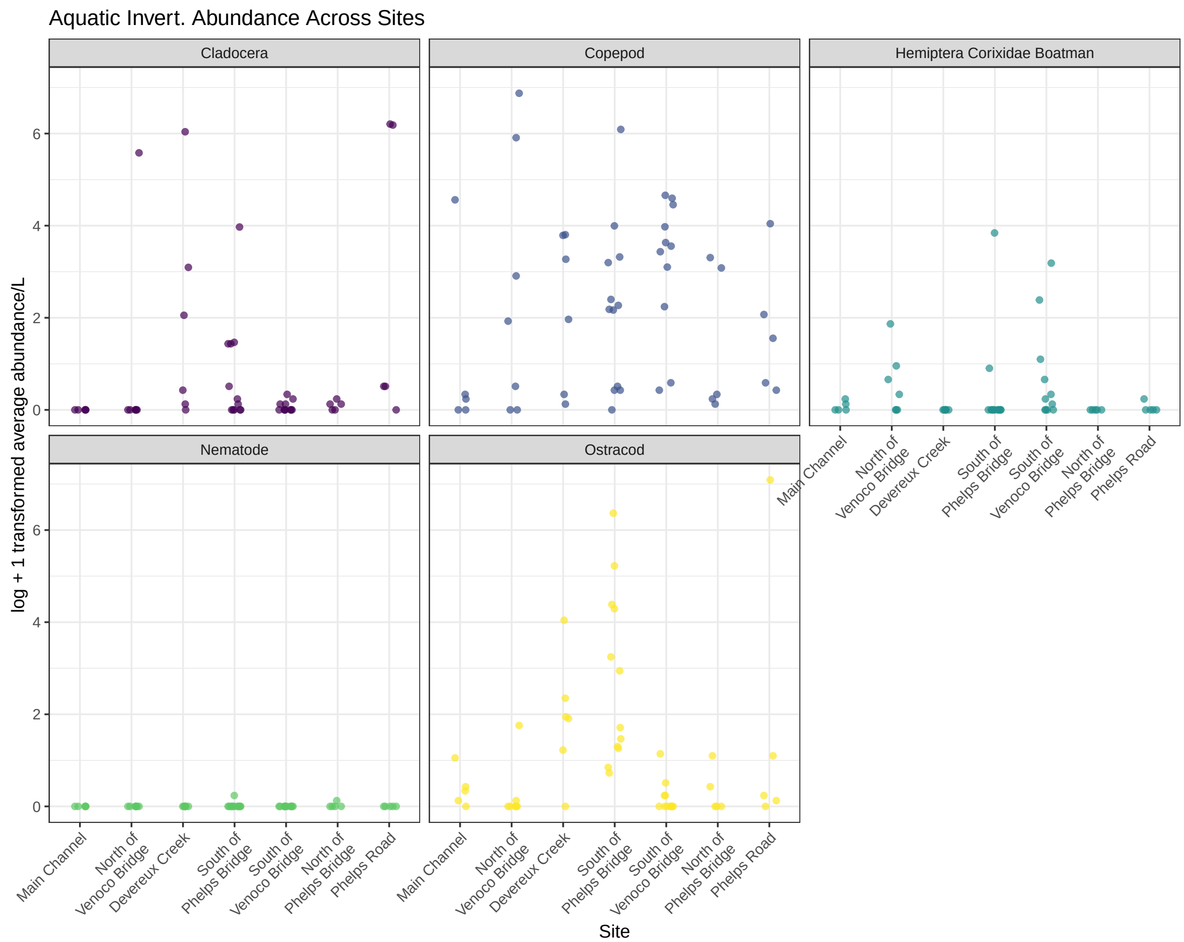


Figure 3: Variance of DO across sites

```
library(ggpointdensity)

# create object aq_ins_clean_fig2 from aq_ins_clean
aq_ins_clean_fig2 <-  aq_ins_clean |>
  # convert taxon to title case
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # filter to include focus taxa
  filter(taxon %in% focus_taxa)

# base layer
```

```r
DO <- ggplot(data = aq_ins_clean_fig2,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_do)) +
  # first layer: boxplot
  geom_boxplot(width = 0.35,
               outlier.shape = NA) +
  # second layer: salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.12,
              shape = 21,
              alpha = 0.6) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # relabelling x- and y-axes
  labs(x = "Site",
       y = "Median DO (mh/L)",
       title = "Site DO levels") +
  # different theme
  theme_bw()

# show plot
DO
```

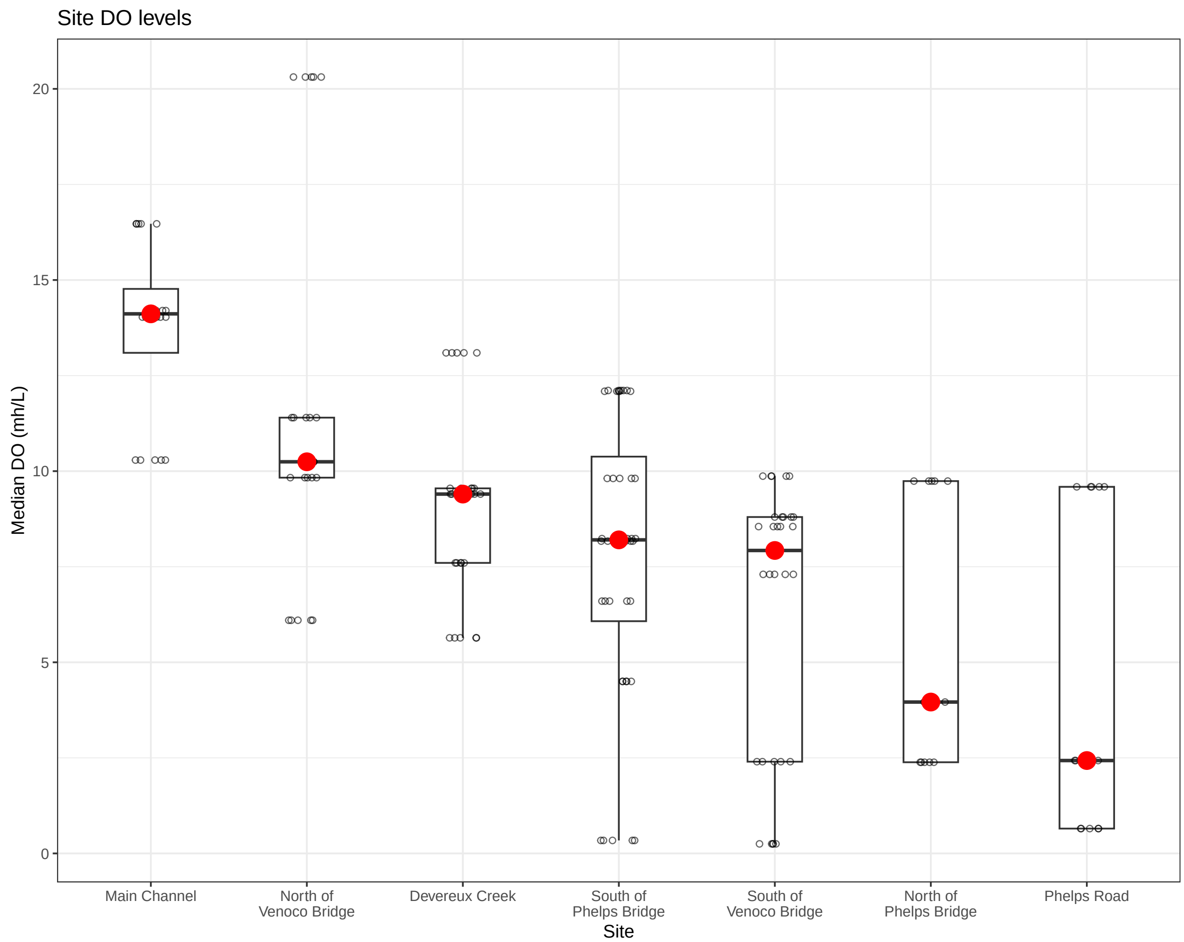
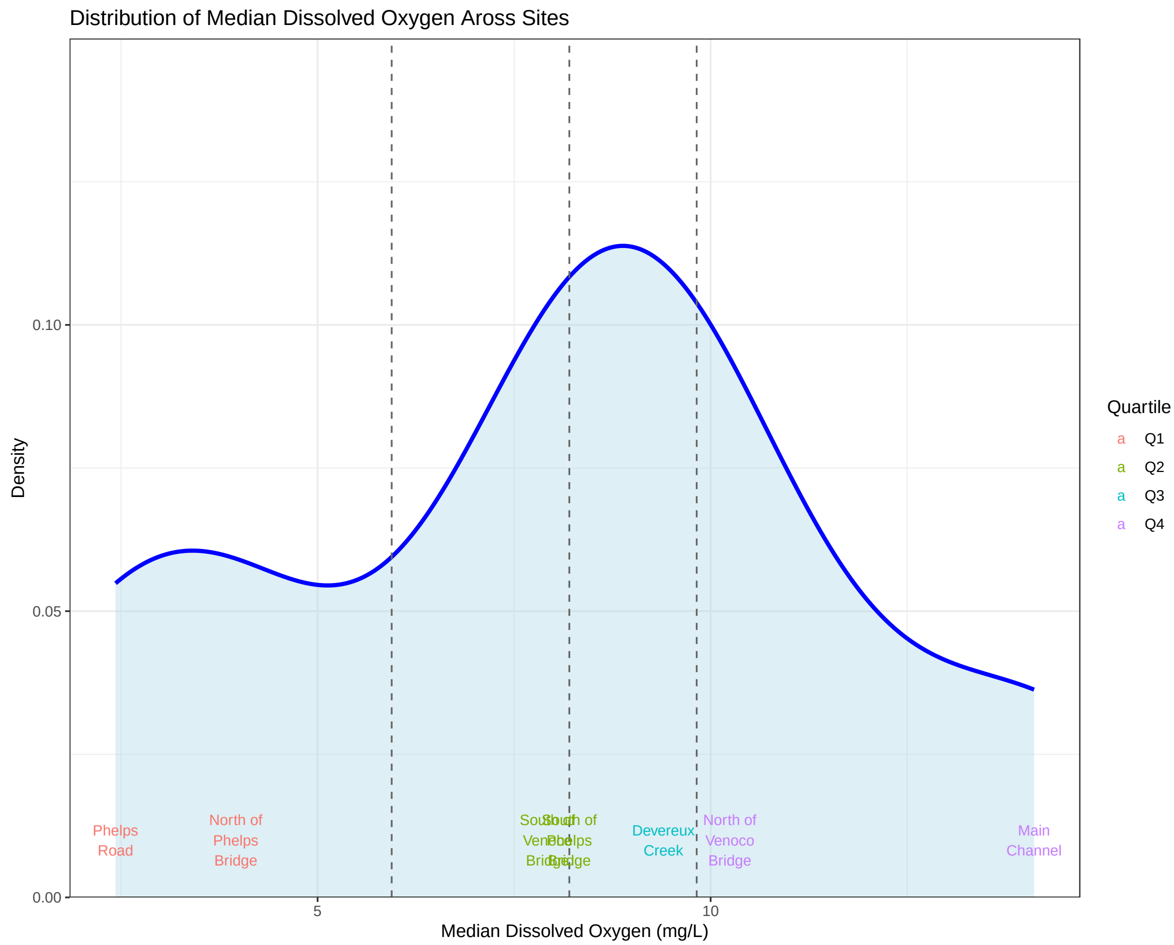


Figure 4: DO Bins Across Sites

```
# create new object to store medians of median do levels
median_fig2 <- aq_ins_clean_fig2 |>
  group_by(site_full) |>
  summarize(median_do = median(med_do, na.rm = TRUE))

# assign quartiles for medians for each site
median_fig2$quartile <- cut(median_fig2$median_do,
                   breaks = quantile(median_fig2$median_do,
                                     probs = seq(0, 1, 0.25)),
                   include.lowest = TRUE,
                   labels = c("Q1","Q2","Q3","Q4"))
```

```r
# boundaries of quartiles
q <- quantile(median_fig2$median_do, probs = c(0.25, 0.5, 0.75))

# distribution plot of DO levels across sites
ggplot(median_fig2, aes(x = median_do)) +
  # density curve
  geom_density(fill = "lightblue",
               alpha = 0.4,
               color = "blue",
               linewidth = 1.2) +
  # vertical quartile lines
  geom_vline(xintercept = q,
             linetype = "dashed",
             color = "gray40") +
  # site labels
  geom_text(aes(y = 0.01,
                label = str_wrap(site_full, width = 10),
                color = quartile),
            hjust = 0.5,
            size = 3) +
  # remove space between bottom of plot and y-axis, set y-axis limits
  scale_y_continuous(expand = c(0,0),
                     limits = c(0, 0.15)) +
  # axes labels and title
  labs(x = "Median Dissolved Oxygen (mg/L)",
       y = "Density",
       title = "Distribution of Median Dissolved Oxygen Aross Sites",
       color = "Quartile") +
  # theme
  theme_bw()
```

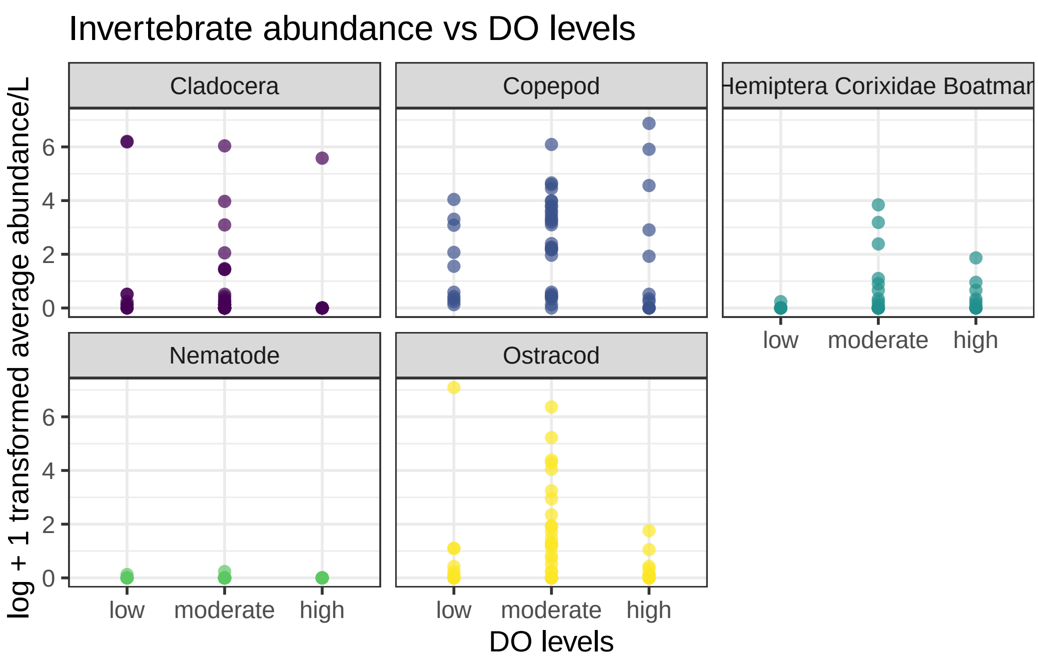


BINS Low DO: Phelps Road and North of Phelps Bridge Moderate DO: South of Venoco Bridge, South of Phelps Bridge, Devereux Creek High DO: North of Venoco Bridge, Main Channel

Figure 5: Invertebrate Abundance in DO Bins

```
aq_ins_clean_bins <-  aq_ins_clean |>
  mutate(bins = case_when(site_full == "Phelps Road" ~ "low",
                          site_full == "North of Phelps Bridge" ~ "low",
                          site_full == "South of Venoco Bridge" ~ "moderate",
                          site_full == "South of Phelps Bridge" ~ "moderate",
                          site_full =="Devereux Creek" ~ "moderate",
                          site_full =="North of Venoco Bridge" ~ "high",
```

```r
                                site_full == "Main Channel" ~ "high")) |>
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  filter(taxon %in% focus_taxa) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  mutate(bins = factor(bins, levels = c("low", "moderate", "high")))


# base layer: ggplot
ins_bins_plot <- ggplot(data = aq_ins_clean_bins,
                        mapping = aes(x = bins, # x-axis: DO bins
                                      y = log_ave_lit, # y-axis: log_ave_lit
                                      color = taxon)) + # color by taxon
  # first layer: points show individual abundance measurements
  geom_point(shape = 16, # shape is circle
             size = 2, # size is 2
             alpha = 0.7) + # opacity is 0.7
  # separate panels for each taxon
  facet_wrap(~ taxon) +
  # use non-default colors for points
  scale_color_viridis_d() +
  labs(x = "DO levels", # relabel x-axis
       y = "log + 1 transformed average abundance/L", # relabel y-axis
       # creates descriptive title
       title = "Invertebrate abundance vs DO levels") +
  # uses non-default theme
  theme_bw() +
  # remove redundant legend
  theme(legend.position = "none")

# shows plot
ins_bins_plot
```



From this figure, it looks like cladocera, copepods, boatmans, and ostracods all prefer moderate levels of DO. Nematodes don't appear to have a preference, but their abundance is low throughout so there might just not be enough to tell.

Abundance over time (might change up this figure or not include)

```
# create object for plotting total abundance over time
abundance_time <- aq_ins_clean |>
  # group by sampling date and taxon
  group_by(date, taxon) |>
  # total abundance across all sites for each date and taxon
  summarize(total_ave_lit = sum(ave_lit, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # create log-transformed total abundance column
  mutate(log_total_ave_lit = log(total_ave_lit + 1)) |>
  # make taxon names easier to read
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # order taxa by maximum total abundance
  mutate(taxon = fct_reorder(taxon, total_ave_lit, .fun = max, .desc = TRUE)) |>
```

```r
  # include main taxon groups of focus
  filter(taxon %in% focus_taxa)


# base layer
abundance_time_plot <- ggplot(data = abundance_time,
                              # x-axis: log_total_ave_lit
                              # fill and color by taxon
                              mapping = aes(x = log_total_ave_lit,
                                            fill = taxon,
                                            color = taxon)) +
  # scaled density makes y-axis consistent across facets
  geom_density(aes(y = after_stat(scaled)),
               alpha = 0.45,
               linewidth = 0.8,
               adjust = 0.8) +
  facet_wrap(~ taxon) + # facet by taxon
  scale_fill_viridis_d() + # non-default colors
  scale_color_viridis_d() + # non-default colors
  labs(x = "log + 1 transformed total average abundance/L across sites",
       y = "Scaled density",
       title = "Distribution of total invertebrate abundance over time") +
  # use clean theme
  theme_bw() +
  # remove redundant legend
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45,
                                   hjust = 1))

# show plot
abundance_time_plot
```

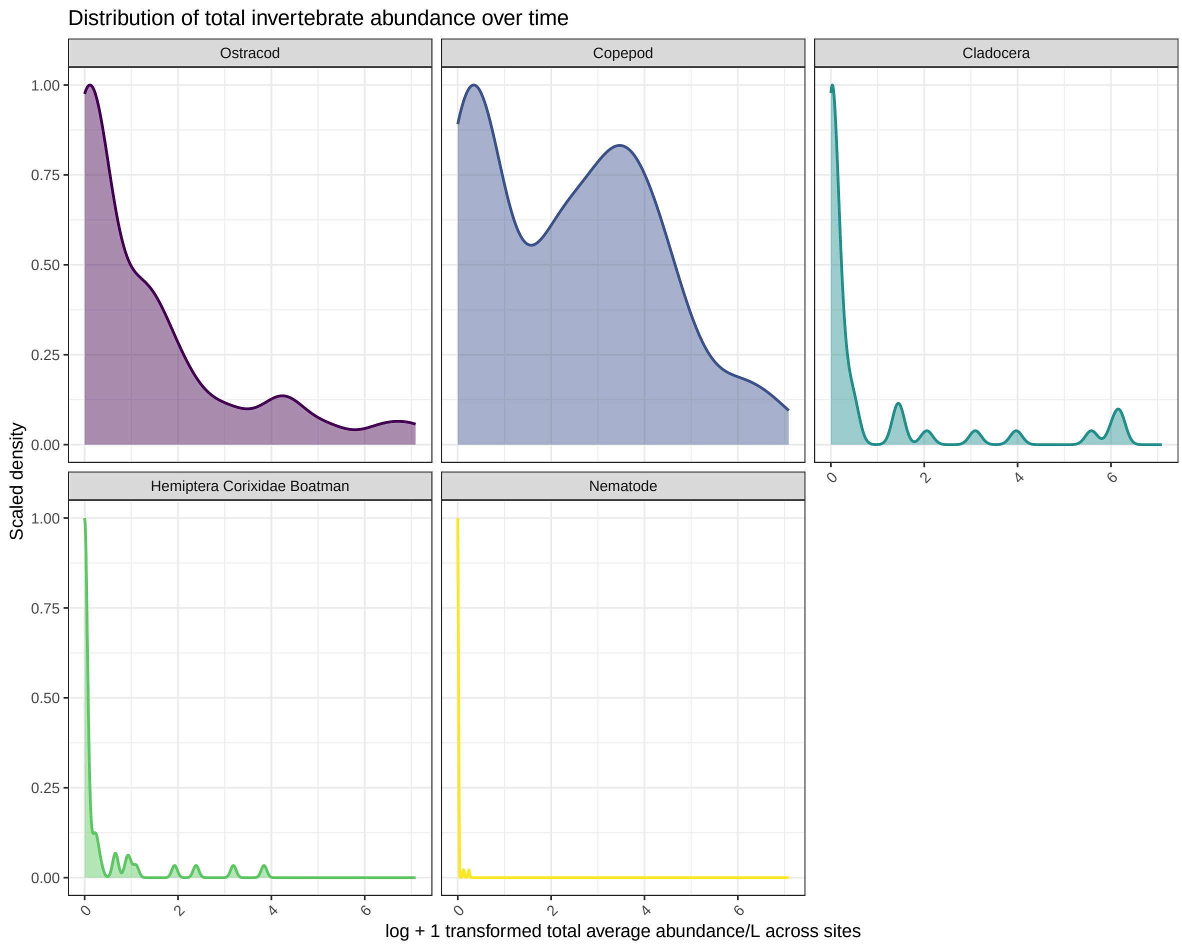
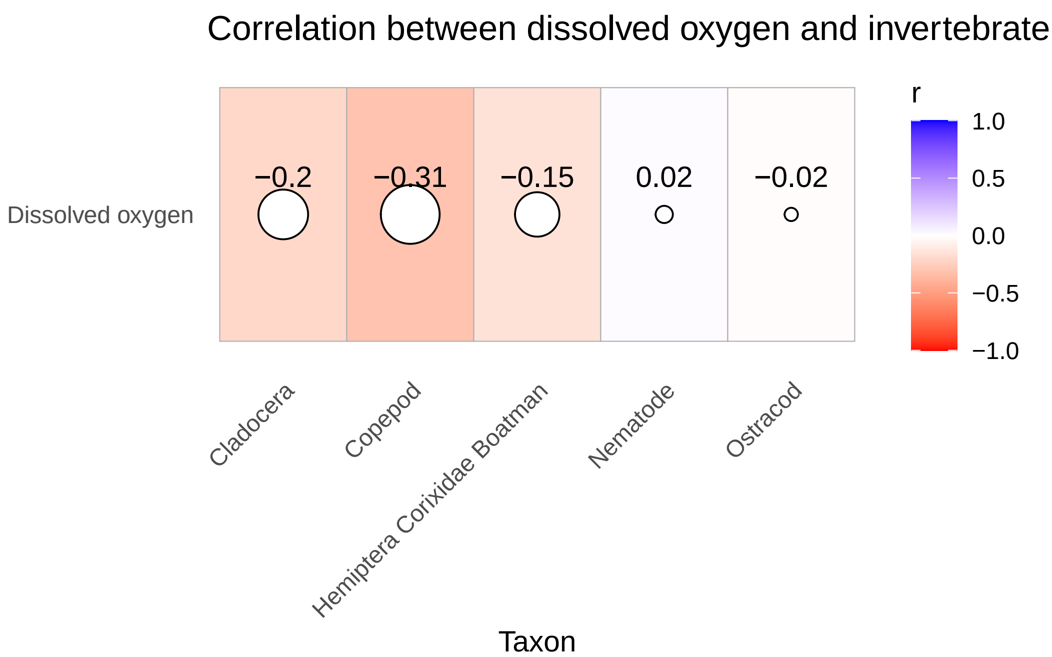


Figure 6: correlation 1

```
# create wide dataset (one column per taxon)
taxa_wide <- DO_taxa |>
  select(date, site, taxon, log_ave_lit, med_do) |>
  pivot_wider(
    names_from = taxon,
    values_from = log_ave_lit
  )

# calculate correlation of each taxon with dissolved oxygen
cor_do <- taxa_wide |>
  select(-date, -site) |>
```

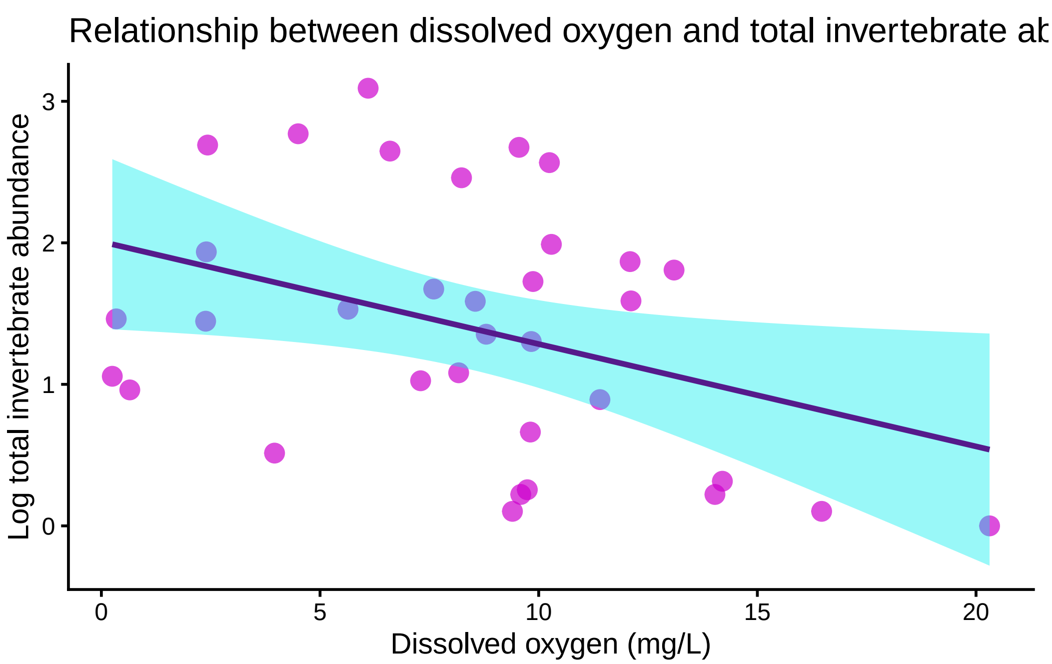
```r
  summarize(across(-med_do, ~ cor(.x, med_do, use = "complete.obs"))) |>
  pivot_longer(
    cols = everything(),
    names_to = "taxon",
    values_to = "correlation"
  )

ggplot(cor_do, aes(x = taxon, y = "Dissolved oxygen", fill = correlation)) +
  geom_tile(color = "grey70") +
  geom_point(aes(size = abs(correlation)), shape = 21, fill = "white") +

  # move labels ABOVE circles
  geom_text(aes(label = round(correlation, 2)),
            nudge_y = 0.15,  # <-- this moves text up
            size = 4) +

  scale_fill_gradient2(
    low = "red",
    mid = "white",
    high = "blue",
    midpoint = 0,
    limits = c(-1, 1),
    name = "r"
  ) +
  scale_size(range = c(2, 10), guide = "none") +
  labs(
    x = "Taxon",
    y = NULL,
    title = "Correlation between dissolved oxygen and invertebrate abundance"
  ) +
  theme_minimal() +
  theme(
    panel.grid = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```



Fiugre 7: Summed abundance across focus taxa vs DO

```
focus_taxa <- c("Cladocera", "Copepod",
                "Hemiptera Corixidae Boatman",
                "Nematode", "Ostracod")

total_abundance <- DO_taxa |>
  filter(taxon %in% focus_taxa) |>
  group_by(date, site, med_do) |>
  summarize(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    log_total_abundance = log10(total_abundance + 1),
    .groups = "drop"
  )
ggplot(total_abundance,
       aes(x = med_do,
           y = log_total_abundance)) +

  geom_point(color = "magenta3",
             size = 3,
```

```r
               alpha = 0.7) +
   geom_smooth(method = "lm",
               se = TRUE,
               color = "purple4",
               fill = "cyan2") +

  labs(
    x = "Dissolved oxygen (mg/L)",
    y = "Log total invertebrate abundance",
    title = "Relationship between dissolved oxygen and total invertebrate abundance"
  ) +

  theme_classic()
```



When taxa are combined, a clearer negative relationship emerges.

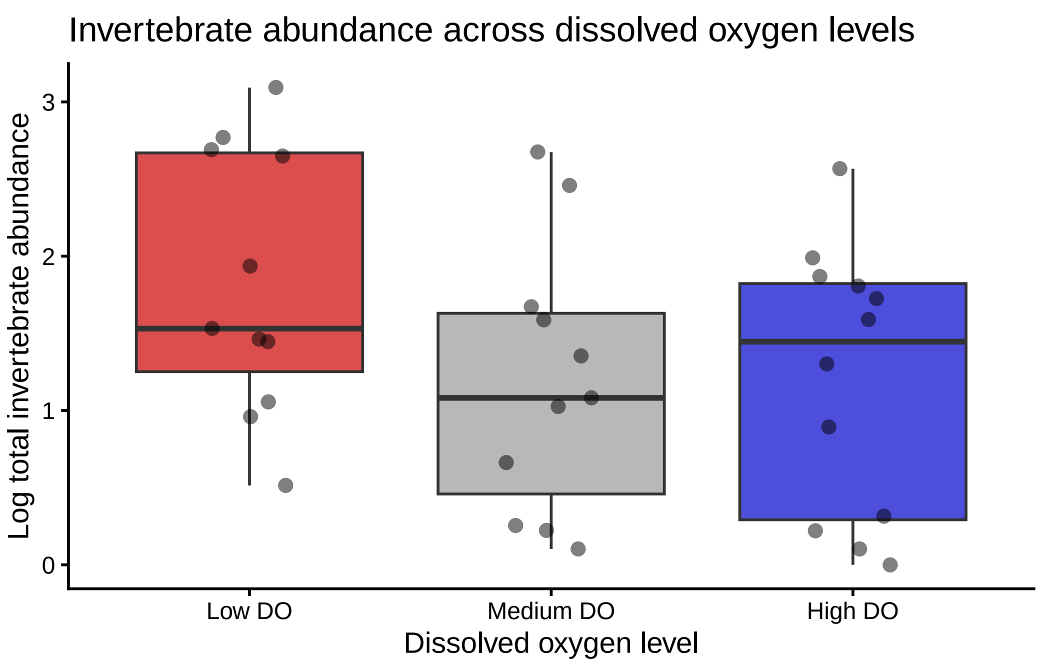
```r
cor.test(
  total_abundance$log_total_abundance,
  total_abundance$med_do,
  method = "pearson"
)
```

```
	Pearson's product-moment correlation

data:  total_abundance$log_total_abundance and total_abundance$med_do
t = -2.2882, df = 32, p-value = 0.02887
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.63289820 -0.04217141
sample estimates:
       cor 
-0.3749894 
```

Binned DO levels

```
total_abundance <- total_abundance |>
  mutate(
    do_category = case_when(
      med_do <= quantile(med_do, 0.33, na.rm = TRUE) ~ "Low DO",
      med_do <= quantile(med_do, 0.66, na.rm = TRUE) ~ "Medium DO",
      TRUE ~ "High DO"
    )
  )
total_abundance$do_category <- factor(
  total_abundance$do_category,
  levels = c("Low DO", "Medium DO", "High DO")
)

ggplot(total_abundance,
       aes(x = do_category,
           y = log_total_abundance,
           fill = do_category)) +

  geom_boxplot(alpha = 0.7,
               outlier.color = "black",
               outlier.size = 2) +

  # optional: add jittered points to show raw data
  geom_jitter(width = 0.15,
              alpha = 0.5,
              size = 2,
              color = "black") +
```

```r
  labs(
    x = "Dissolved oxygen level",
    y = "Log total invertebrate abundance",
    title = "Invertebrate abundance across dissolved oxygen levels"
  ) +

  scale_fill_manual(values = c(
    "Low DO" = "red3",
    "Medium DO" = "grey60",
    "High DO" = "blue3"
  )) +

  theme_classic() +
  theme(legend.position = "none")
```



Community abundance varies continuously rather than discretely across DO levels.

```r
# Pearson correlation for continuous DO and total community abundance
cor.test(total_abundance$log_total_abundance,
         total_abundance$med_do,
         method = "pearson")
```

```
	Pearson's product-moment correlation

data:  total_abundance$log_total_abundance and total_abundance$med_do
t = -2.2882, df = 32, p-value = 0.02887
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.63289820 -0.04217141
sample estimates:
       cor 
-0.3749894 
```

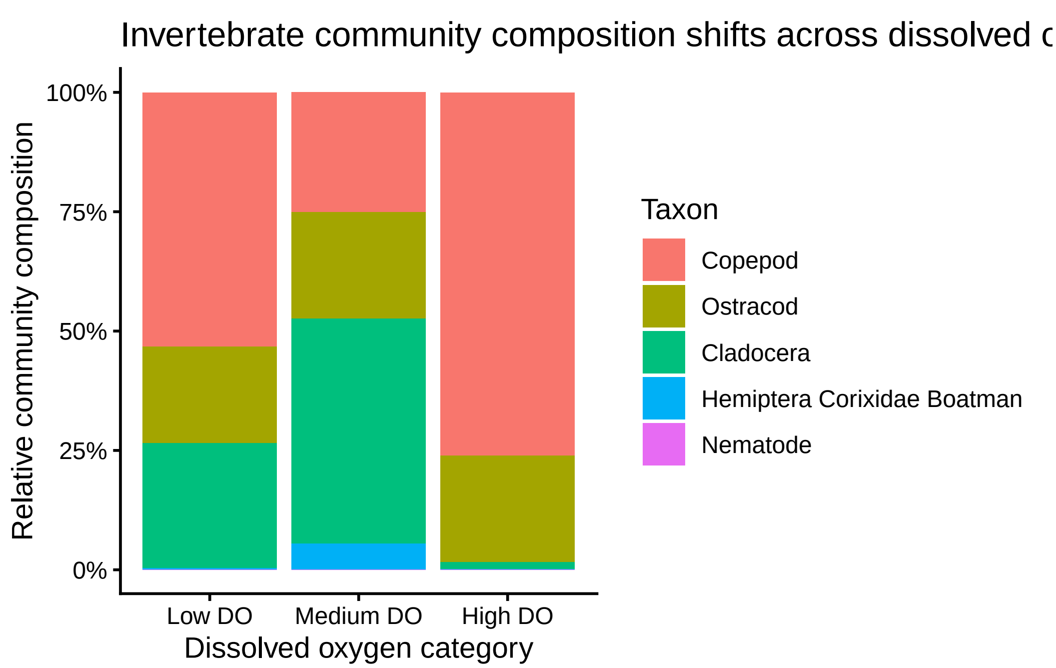
```
# Kruskal-Wallis test comparing abundance across DO categories
kruskal.test(log_total_abundance ~ do_category,
             data = total_abundance)
```

```
	Kruskal-Wallis rank sum test

data:  log_total_abundance by do_category
Kruskal-Wallis chi-squared = 3.2212, df = 2, p-value = 0.1998
```

Figure 8: Composition plot

```
composition_plot <- DO_taxa |>
  mutate(
    do_category = case_when(
      med_do <= quantile(med_do, 0.33, na.rm = TRUE) ~ "Low DO",
      med_do <= quantile(med_do, 0.66, na.rm = TRUE) ~ "Medium DO",
      TRUE ~ "High DO"
    ),

    # reorder factor levels
    do_category = factor(do_category,
                         levels = c("Low DO",
                                    "Medium DO",
                                    "High DO"))
  ) |>
  group_by(do_category, taxon) |>
```

```
  summarize(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  ) |>
  ggplot(aes(x = do_category,
             y = total_abundance,
             fill = taxon)) +

  geom_col(position = "fill") +

  scale_y_continuous(labels = scales::percent_format()) +

  labs(
    x = "Dissolved oxygen category",
    y = "Relative community composition",
    fill = "Taxon",
    title = "Invertebrate community composition shifts across dissolved oxygen levels"
  ) +

  theme_classic()

composition_plot
```



Medium DO sites show a more even distribution of taxa and greater relative abundance of Cladocera and Ostracods, while low and high DO conditions are dominated primarily by Copepods.